SAP JOULE ENABLEMENT

Runbook: getting a Joule-enabled SAP environment up and running

Covers SAP-managed and customer-managed provisioning, SAP BTP and identity foundations, every major SAP product integration, Joule Studio, document grounding, Microsoft 365 Copilot, testing, operations and troubleshooting.

Document	Detail
Version	1.1 (adds reviewed field-practice notes to 1.0)
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Audience	SAP BTP administrators, Basis and identity teams, SAP product administrators, project leads
Status	Researched from SAP Help, SAP Learning, SAP Architecture Center and SAP Community. Items tagged VERIFY must be checked against the live SAP page before use.
Companion file	`research-notes.md` in the same folder: every finding with its source and confidence level
Change in 1.1	Field-practice additions from a peer review (Antigravity), checked against the original research and tagged where unconfirmed. See section 1.4 and Appendices E and F.

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1. How to use this runbook

Joule is SAP's generative-AI assistant. It sits on SAP Business Technology Platform (BTP), signs users in through SAP Cloud Identity Services, finds the right apps through SAP Build Work Zone, and acts in each connected SAP product with that user's own permissions.

There are two ways to switch it on, and the right one depends on your products and how much control you need. Read sections 2 to 5 first; they apply to everyone. Then follow either section 6 (SAP-managed) or section 7 (customer-managed), and only the product chapters in section 8 that you own.

1.1 Reading path

If you are...	Read
Deciding whether and how to start	2 Choose your path · 3 Licences · 4 Landscape design
Starting fresh with SAP cloud products	5 People and access · 6 SAP-managed provisioning · 11 Testing
Setting it up yourself on SAP BTP	5 · 7 Common foundation · the product chapters in 8 · 11 · 12
Adding custom skills, documents or Microsoft 365	9 Extending Joule
Enabling consultants or ABAP developers	10 Joule for Consultants and Joule for Developers
Fixing a problem	13 Troubleshooting · Appendix B SAP Notes
Preparing go-live	14 Security · 15 Master go-live checklist

1.2 Conventions

- **Bold** marks names you click or select in a cockpit or app. `Monospace` marks technical names, IDs, paths and URLs.
- The amber tag VERIFY marks a detail that came from a single or non-SAP source, or where sources disagree. Check it against the SAP Help page named in the step before you rely on it.
- SAP releases Joule changes frequently (SAP Help pages for Joule were updated through October 2026). Treat SAP Help and the SAP Notes in Appendix B as the final word.

> **Why the verification tags**
> This runbook was compiled from search extracts of SAP Help, SAP Learning, the SAP Architecture Center and SAP Community, because direct access to those sites was blocked in the research environment. Plan names, role names and menu paths change between releases, so confirm the tagged items during your own setup.

1.3 How a Joule request travels

Joule is not a separate chatbot. Every answer depends on an unbroken chain from the signed-in person to the back-end system. When something fails, walk the chain in order: the first broken link is almost always the cause.

#	Link	What must be true	Typical symptom when broken
1	Identity	The user signs in through the shared IAS tenant and carries the same Global User ID everywhere	Login loops, blank Joule window
2	Roles	BTP role collection (via IAS group) plus the product's business roles or permissions	Joule icon missing, or works for admins only

#	Link	What must be true	Typical symptom when broken
3	Object authorisation	The user may see the company code, plant, legal entity or record asked about	Answers without data, or "no permission"
4	System registration	The product system is registered and in the Joule formation	System missing from the booster
5	Content exposure	The product publishes its apps and roles (for example CDM3 content in Private Edition)	Joule cannot find an app the user has
6	Navigation	Work Zone has the content and the user's roles (IPS keyed on Global User ID)	"Could not resolve navigation target"
7	Execution	Calls reach the back end as the user (principal propagation), not a technical user	Empty results or HTTP 403 on transactional questions

1.4 What's new in version 1.1

Version 1.1 adds field-practice notes from a peer review of version 1.0. Each addition was checked against the original research: confirmed or plausible items are folded into the steps below (tagged VERIFY where SAP documentation was not seen), specific values without a source are parked in Appendix E, and suggestions that conflict with SAP sources or carry risk are listed in Appendix F with the reason.

- New: request chain and diagnostic order (1.3).
- Identity: send IAS groups in the token, disable SAP ID Service for business-user logon, key IPS on the Global User ID (7.4, 7.6, 7.10).
- BTP: registered versus discovered systems, one formation per system, System Landscape Administrator, booster retry guidance (4.2, 5.1, 7.7, 7.8).
- Roles: add the Work Zone launchpad roles to the Joule role collections; map through IAS groups (7.9).
- S/4HANA: Joule plugin business catalog (8.1); complete principal-propagation steps for Private Edition (8.2).
- Operations: certificate-expiry register, troubleshooting extended from 14 to 26 entries (12.4, 13).
- Extensions: Joule Studio destination property, grounding data hygiene, one Entra app registration per tier (9).

2. Choose your path

2.1 The two provisioning models

	SAP-managed provisioning	Customer-managed provisioning
What it is	You request Joule in SAP for Me. SAP sets up BTP, identity and the integrations, and runs lifecycle operations.	You set up the BTP subaccount, SAP Cloud Identity Services trust, SAP Build Work Zone and run the Joule booster yourself.
Status	Announced at SAP Sapphire 2026 as the standard activation path for new Joule deployments.	Fully supported for existing deployments; SAP offers migration help towards SAP-managed.
Effort	Low: guided steps in SAP for Me, then product-side settings.	Medium to high: BTP, identity, Work Zone, destinations, roles.
Control	Less: SAP owns the BTP side.	Full: your own subaccount, roles, destinations, extensions.
Best for	New Joule customers on SAP cloud products who want the fastest route.	Landscapes that need Joule Studio, document grounding, Microsoft 365 Copilot, S/4HANA Private Edition specifics, or full control over BTP.
Where	Section 6	Section 7, then section 8

Mixing models
SAP documents that running both models in parallel is possible, but recommends one model per landscape for a consistent experience (stated for Joule Studio). Decide early, because the BTP subaccount layout differs.

Practical consequences of each model
SAP-managed: SAP owns the BTP side, so you do not administer its subaccount; you cannot add your own destinations or BTP services there. VERIFY Check that each extension you plan (Joule Studio, document grounding, Microsoft 365, Private Edition) is offered for SAP-managed before you choose it.

Mixing models for different products on the same IAS tenant can produce two separate Joule instances, so users see separate chats and cross-product requests do not work. VERIFY Plan one model per tier.

2.2 Path by product

Product	Typical path	Key extra requirement	Chapter
S/4HANA Cloud Public Edition	SAP-managed or booster	Communication arrangements created by the booster; Content Security Policy; Joule plugin catalog `SAP_CORE_BC_UI_JOU_PC` VERIFY	8.1
S/4HANA Cloud Private Edition (RISE)	Customer-managed (booster or Cloud Integration Automation service)	Release 2021 or later (2023 FPS01 recommended), Cloud Connector, principal propagation, content exposure	8.2
S/4HANA on-premise / ECC	Hybrid, subject to SAP eligibility	Cloud commitment conditions announced 2026 VERIFY	8.3
SAP SuccessFactors	SAP-managed or booster	Role-based permission "Access to Joule"	8.4

Product	Typical path	Key extra requirement	Chapter
SAP Concur	Booster	Concur domain in security settings	8.5
SAP Ariba	Booster	Ariba OAuth application, realm data centre	8.6
SAP Analytics Cloud	Booster	Work Zone mandatory since Joule 2506; "Just Ask" enabled	8.7
SAP Datasphere	Booster	Role DW Joule Consumer; OAuth client	8.8
SAP IBP	Booster	Catalog SAP_CORE_BC_UI_JOU_PC	8.9
Signavio, Fieldglass, IPD, CX and others	Per SAP Help integration page	See each product guide	8.10

3. Licences and commercial prerequisites

> **Confirm with your SAP account team**
> Commercial terms for Joule changed in July 2025 and again in 2026. Use this section to prepare the conversation, not as a price list.

3.1 What you need

Item	What it covers	How to check
Licence for each SAP product	Joule only works in products you license (for example S/4HANA Cloud, SuccessFactors, Concur).	SAP for Me → Portfolio & Products
Joule Base	Included with qualifying SAP cloud subscriptions. Covers the core assistant (navigation, informational and transactional scenarios).	SAP for Me → Portfolio & Products → Business AI
AI Units (Joule Premium)	Consumption currency for premium capabilities such as Joule Agents and assistants. Purchased and activated in SAP for Me; unused units may expire at the end of the term. VERIFY	SAP for Me; order form
Onboarding SKU	SAP's onboarding guide lists "purchased the AI Unit SKU `8019164` or Joule Base" as a prerequisite. VERIFY	Order form
SAP BTP enterprise global account	Needed for customer-managed provisioning, Joule Studio, document grounding. Contract types cited: CPEA, BTPEA or subscription.	BTP cockpit → global account
SAP Build licence (Joule Studio)	SAP Build entitlement plus SAP Build developer licence to build, test and deploy skills and agents; Joule entitlement to run them in production.	BTP entitlements
Microsoft 365 Copilot licence	Only for the Microsoft 365 Copilot / Teams integration where Copilot skills are used.	Microsoft 365 admin centre

3.2 Questions to settle with SAP

- Which of our products are entitled to Joule Base today, and in which data centres?
- How many AI Units do we need for the agents we plan, and when do they expire?
- Does our contract allow SAP-managed provisioning, or do we stay customer-managed?
- For on-premise S/4HANA or ECC: are we eligible for the 2026 hybrid Joule offering, and under which cloud commitment? VERIFY
- Are there current promotions (for example the 2026 free run of custom Joule agents for SAP Build customers)?

4. Landscape and architecture design

4.1 How Joule fits together

Component	Role in Joule
Joule (BTP service, technical name `das-application`)	The assistant itself: dialogue, intent recognition, generative answers, scenario routing.
SAP Cloud Identity Services: Identity Authentication (IAS)	Single sign-on for Joule and every connected product. One IAS tenant per landscape tier.
SAP Cloud Identity Services: Identity Provisioning (IPS)	Synchronises users, groups and roles (SCIM 2.0, including `userUuid`) to Work Zone and products.
SAP Build Work Zone, standard edition	Navigation service: resolves "open the app for..." requests to the right app the user is allowed to use.
Formation (BTP System Landscape)	Groups Joule with the product systems it integrates; created by the booster.
Destinations	How Joule and Work Zone reach each product system.
Cloud Connector	Secure outbound tunnel to private systems (S/4HANA Private Edition, on-premise).
Generative AI hub in SAP AI Core	SAP-managed model routing behind Joule; not configured by customers.
XSUAA (authorisation service of the subaccount)	Issues the tokens that carry the user's role-collection scopes; it only sees the groups IAS sends.
SAP HANA Cloud Vector Engine	Stores document embeddings for document grounding (9.1).
Cloud Integration Automation service (CIAS)	Guided automation of multi-step setups (Private Edition, document grounding).

4.2 Design rules

- **One IAS tenant per tier.** Joule and every product integrated with it must authenticate against the same SAP Cloud Identity Services tenant. Different IAS tenants mean separate Joule instances.
- **One Global User ID per person.** Each user must carry the same Global User ID (`user_uuid`) in every connected product. This is the most common cause of failed sessions.
- **Two tiers.** SAP's landscape recommendation is a non-production Joule (on the **Joule Preview** landscape) for sandbox, development and test, and a production Joule. Older guidance shows three stages.
- **Same subaccount.** Joule and SAP Build Work Zone, standard edition live in the same subaccount, with the Work Zone destinations. Joule talks to only one Work Zone instance; SAP recommends reusing an existing Work Zone.
- **Same region.** Keep the subaccount, IAS tenant and Work Zone in the same region, with the Cloud Foundry environment enabled.
- **Supported regions differ by product.** Examples seen: Joule for Consultants EU10, EU11, EU20, US10, AP10, JP10; Joule Studio AP10, EU10, EU30, EU20, US10, US21; Joule Preview entitlements only on EU10 or EU30. Check the SAP Discovery Center before creating subaccounts. VERIFY
- **Separate subaccounts where SAP requires them.** Joule for Consultants needs its own subaccount; some SAC guides use a separate subaccount for identity provisioning.

- **One formation per product system.** A product system (for example one S/4HANA tenant or one SuccessFactors company) can belong to only one Joule formation at a time; adding it to a second formation fails. VERIFY
- **Keep Joule close to the back end.** Put the Joule subaccount in a region near your main product systems to keep response times low.

4.3 Recommended layout

Tier	Global account / subaccount	IAS tenant	Connected systems
Non-production	Enterprise global account → subaccount `joule-nonprod` (Cloud Foundry, supported region), Joule Preview entitlement where available	Test IAS tenant	Sandbox, development and quality systems of each product
Production	Same or separate global account → subaccount `joule-prod`	Productive IAS tenant	Productive systems of each product
Optional	Separate subaccounts for Joule for Consultants and for Joule Studio development	Same tier's IAS	As needed

5. People, access and readiness

5.1 Who you need

Role	Needed for	Access
SAP for Me administrator	SAP-managed provisioning, AI Unit activation, product packages	Cloud Administrator or Super Administrator in SAP for Me
BTP global account administrator	Entitlements, boosters, System Landscape, formations	Global Account Administrator; also System Landscape Administrator if your global account separates it VERIFY
BTP subaccount administrator	Subscriptions, trust, role collections, destinations	Subaccount Administrator
Identity administrator	IAS applications, trusted domains, user attributes, IPS jobs, Entra ID federation	IAS administrator
Product administrator(s)	Product-side settings (communication arrangements, RBP, catalogs, OAuth clients)	Product admin roles
Basis administrator (Private Edition / on-premise)	SICF services, content exposure, certificate mapping, Cloud Connector	Basis authorisations
Network and security	Cloud Connector, allowlists, Content Security Policy, browser policies	Firewall and proxy owners
Business key users	Acceptance testing in each product	Test users with real roles

5.2 Readiness checklist

☐	Item	Owner	Ref
☐	Product licences and Joule Base / AI Units confirmed	Project lead	3
☐	Provisioning model chosen (SAP-managed or customer-managed)	Architect	2
☐	Enterprise BTP global account available	BTP admin	3
☐	IAS tenants identified per tier and already used by the products	Identity admin	4.2
☐	Every user has a Global User ID in IAS	Identity admin	7.6
☐	Supported region chosen for each subaccount	Architect	4.2
☐	Existing Work Zone, standard edition site identified for reuse	BTP admin	4.2
☐	Cloud Connector available (Private Edition / on-premise only)	Basis	8.2
☐	Test users per product with representative roles	Key users	11
☐	Data-protection review started	Security / DPO	14
☐	IAS application sends user groups in the token	Identity admin	7.6
☐	Browser third-party cookie policy checked (Joule runs embedded in the product UI)	Workplace IT	8.4

☐	Item	Owner	Ref
☐	Certificate register started for Cloud Connector and trust certificates	Basis	12.4

6. Path A: SAP-managed provisioning

Use this path for new Joule deployments where SAP offers it for your products. SAP configures SAP BTP, Cloud Identity Services and the product integrations; you complete the product-side settings and assign users.

6.1 Steps

1. Confirm prerequisites: licensed SAP cloud product that supports Joule, in a supported region; Joule Base or AI Units purchased; all products on one IAS tenant per tier.
2. Sign in to **SAP for Me** with the Cloud Administrator or Super Administrator role.
3. Accept the SAP AI terms if prompted (some contracts use a zero-cost line item for this). VERIFY
4. Open **Portfolio & Products → Business AI** and find the Joule package. Choose **Ready for Activation**.
5. Open **Portfolio & Products → My Product Packages**, select the package and choose **Start Provisioning**.
6. Monitor progress on the **Resources** tab until provisioning completes.
7. Complete the product-side settings for each product (the "Product side" steps in section 8), for example role-based permissions in SuccessFactors or business catalogs in S/4HANA Cloud and IBP.
8. Assign users through the product's roles and the IAS groups SAP set up.
9. Run the tests in section 11.

Testing levels

SAP Community describes two SAP-managed test approaches: **Level 2** (recommended, customer-like end-to-end provisioning through SAP for Me) and **Level 0** (a quick background provisioning for preliminary validation when Level 2 is temporarily unavailable).

What you still own

User and role assignment, product-side configuration (for example the Joule plugin catalog in S/4HANA Cloud or the "Access to Joule" permission in SuccessFactors), Content Security Policy and trusted domains on the product side, testing, change management and AI Unit monitoring.

Extensions such as Joule Studio may have their own SAP-managed option (SAP Help: SAP-Managed Provisioning for Joule Studio).

7. Path B: customer-managed common foundation

These steps build the SAP BTP and identity foundation once per tier. Every product chapter in section 8 starts from here. SAP Help: *Customer-Managed Provisioning* and *Prerequisites for Customer-Managed Provisioning*.

7.1 Prepare the global account and subaccount

1. In the BTP cockpit, open the enterprise global account. Confirm the Joule entitlement is present under **Entitlements**.
2. Create or choose the subaccount (for example `joule-nonprod`) in a supported region, ideally the one that already hosts your Work Zone site.
3. Enable the **Cloud Foundry** environment and create a space.
4. Confirm the subaccount, the IAS tenant and Work Zone are in the same region.

7.2 Assign entitlements

Under **Entitlements → Entity Assignments**, add to the subaccount:

Service	Plan	Notes
Joule (`das-application`)	`foundation` in most guides; some newer guides show `standard` or DAS Standard / DAS Premium VERIFY	Joule Preview entitlement for non-production where offered (EU10 / EU30)
SAP Build Work Zone, standard edition	`foundation` (included in every global account); some SuccessFactors guides use `standard` VERIFY	Reuse an existing Work Zone subscription if you have one
SAP Cloud Identity Services	per your contract (guides cite `default`, `connectivity` or an additional-tenant plan) VERIFY	Only if the IAS tenant is not yet linked to the global account
Extensions (optional)	Document Grounding `data-manager`; SAP Build Process Automation `build_default`	See section 9

7.3 Subscribe to SAP Build Work Zone, standard edition

1. In **Instances and Subscriptions**, create the subscription to SAP Build Work Zone, standard edition (skip if it exists).
2. Assign the role collection `Launchpad_Admin` to the administrator.
3. Open the Work Zone site manager once to confirm it loads. If the booster later reports that Work Zone is not registered (KBA `3721340`), create a site in the site manager, publish it, and resume. VERIFY

7.4 Establish trust with Identity Authentication (OpenID Connect)

1. In the subaccount, open **Security → Trust Configuration → Establish Trust** and select the IAS tenant that your SAP products use.
2. Confirm the trust type is **OpenID Connect**. Joule does not support SAML for this trust.
3. Under **Security → Trust Configuration**, open the **Default Identity Provider** (SAP ID Service) and clear **Available for User Logon** for business users, so everyone signs in through your IAS tenant. If users can still sign in through SAP ID Service, their tokens come from the wrong origin and Joule refuses them.

4. Keep at least one administrator able to sign in through SAP ID Service, or a tested IAS admin, before you switch it off.

7.5 Configure trusted domains

1. In IAS, add each product domain (for example the S/4HANA, SuccessFactors, Concur or Datasphere host) as a **trusted domain**.
2. In the BTP subaccount, add the product hosts under **Security → Settings → Trusted Domains**.
3. In each product, allow the Joule URL in its Content Security Policy or allowlist (for example **Manage Content Security Policy** in S/4HANA Cloud Public Edition).
4. In the IAS application for the subaccount, check the OpenID Connect **redirect URIs** include the subaccount's login callback (`https://<subdomain>.authentication.<region>.hana.ondemand.com/login/callback`); the trust setup normally adds it.

7.6 Configure user attributes (Global User ID)

1. In IAS, open the application created for the subaccount trust.
2. Set the **subject name identifier** to the **Global User ID** from the Identity Directory.
3. Under assertion attributes, map `user_uuid` to the Global User ID. If `user_uuid` is not listed, add it as a custom attribute sourced from the Identity Directory.
4. Set the Work Zone application's subject name identifier to the Global User ID as well, so the navigation service can match users.
5. Check that every user exists in the IAS user store with a Global User ID; Joule cannot recognise users without one.
6. Make sure the IAS application **sends the user's groups** in the token (assertion attribute `groups`). Role collections mapped to IAS groups (7.9) only work if the groups arrive; otherwise users end up with no role collections.
7. Optionally also map `user_name` (login name) and `mail` (e-mail) so names and addresses show correctly; never use e-mail as the subject name identifier.

7.7 Register product systems

Products that need registration (for example S/4HANA Cloud Public Edition and IBP) must appear in the global account under **System Landscape → Systems** before the booster can integrate them. Many SAP cloud products appear automatically; others need a registration token configured in the product. Follow the product chapter in section 8. VERIFY

- A system that shows only as **discovered** (found automatically through your customer number) may not be selectable; open it and choose **Register**, then complete the registration in the product. VERIFY
- Only registered systems appear in the booster and in **Include Systems**.

7.8 Run the Joule booster

1. In the global account, open **Boosters**, search for **Setting Up Joule** (shown as "Joule" in some releases) and choose **Start**.
2. Let the booster check entitlements, authorisations and the IAS tenant. Fix anything it reports (see section 13: missing Work Zone registration, missing IAS tenant).
3. Select the subaccount, the integration scenarios, and the SAP systems to integrate. Give the formation a name, for example `joule-nonprod`.

4. Finish the booster. It creates the Joule subscription, destinations and the formation, and (for S/4HANA Cloud Public Edition) communication arrangements in the background.
5. Check **System Landscape → Formations**: the formation should show status **Ready** and list your systems.
6. To add another product later, open the formation and use **Include Systems**, or run the booster again for that system.

The booster does not create role collections
Create them yourself in 7.9, or users will not see Joule even though the setup succeeded.

If the booster fails
Fix the reported cause first (section 13), then check whether a formation was already created before you start a new run: a second run can leave a half-built formation, and a system can belong to only one formation. Resume or reuse the existing run or formation where the cockpit offers it. VERIFY

Do not rename or hand-edit the destinations the booster creates; change settings through the formation or by re-running the booster.

7.9 Create role collections and assign users

Role collection (suggested name)	Roles from application `das-application`	Assign to
`JOULE_END_USER`	`end_user`, plus `Launchpad_User` from Work Zone if your users need the launchpad there	Every Joule user (map to an IAS group such as `ALL_EMPLOYEES`)
`JOULE_ADMIN`	`tenant_admin`, `analytics_admin` and other admin roles listed in the subaccount, plus `Launchpad_Admin`	Joule administrators (a small named IAS group)
`JOULE_EXTENSIBILITY` (optional)	`extensibility_developer`	Joule Studio developers

1. In the subaccount, open **Security → Role Collections → Create**.
2. Add the roles above. Role names can differ slightly per data centre; pick them from the `das-application` list. VERIFY
3. Map each role collection to an IAS user group (**Trust Configuration → Role Collection Mappings**) or assign users directly under **Security → Users**.
4. Prefer least privilege: do not hand out admin role collections to test end users.
5. Prefer IAS group mappings over assigning individual users in the cockpit: they are easier to audit and leavers lose access when IAS removes them.

7.10 Configure Identity Provisioning

1. In the Identity Provisioning Service, create a source system (IAS or the product) and a target system **SAP Build Work Zone** for the subaccount.
2. Map users, groups and roles; include `userUuid` (Global User ID), and key users on that immutable ID, not on e-mail. Keying on e-mail creates duplicate users when addresses change and breaks the navigation match between Joule and Work Zone.
3. Run the job once, then schedule it (for example hourly) so role changes reach Work Zone.

7.11 Foundation checklist

☐	Item	Owner	Ref
☐	Subaccount in supported region, Cloud Foundry enabled	BTP admin	7.1
☐	Joule, Work Zone and Cloud Identity Services entitlements assigned	BTP admin	7.2
☐	Work Zone, standard edition subscribed; Launchpad_Admin assigned	BTP admin	7.3
☐	OIDC trust to the shared IAS tenant	Identity admin	7.4
☐	Trusted domains in IAS, BTP and product CSP	Identity / product admin	7.5
☐	Global User ID as subject name identifier; user_uuid mapped	Identity admin	7.6
☐	Product systems registered in System Landscape	BTP admin	7.7
☐	Booster finished; formation status Ready	BTP admin	7.8
☐	Role collections created and mapped to IAS groups	BTP admin	7.9
☐	IPS job to Work Zone scheduled	Identity admin	7.10

8. Product integrations

Each chapter assumes the foundation in section 7 (or SAP-managed provisioning in section 6). SAP Help hosts one integration page per product under *Integrating Joule with SAP*.

8.1 SAP S/4HANA Cloud Public Edition

Prerequisites

- S/4HANA Cloud Public Edition tenant using the same IAS tenant as the BTP subaccount trust.
- Administrator with access to communication management and Content Security Policy apps; `Launchpad_Admin` in BTP.
- Only one S/4HANA Cloud Public Edition tenant can currently connect to a Joule tenant (stated for Joule for Developers). VERIFY

Steps

1. Register the S/4HANA Cloud system in **System Landscape** (if it is not already listed) and complete the registration in S/4HANA. VERIFY
2. Run the booster (7.8) and select the S/4HANA Cloud system. In the background it creates communication arrangements `SAP_COM_0882` and `SAP_COM_0647`.
3. In S/4HANA, open communication arrangement `SAP_COM_0647` and set the role exposure to **ALL**. If you keep **SELECTED**, maintain the filter `s4hana.cloud.roles.filter – cFLGExposure eq true`.
4. Create and activate `SAP_COM_0193` manually so users and roles are provisioned to Work Zone (with IPS, 7.10).
5. In **Manage Content Security Policy**, add the Joule application URL to the trusted sites.
6. Add the S/4HANA tenant URL to the BTP subaccount trusted domains.
7. In Work Zone, refresh the S/4HANA content provider so apps and roles are available to the navigation service.
8. Add business catalog **User Interface – Joule Plugin** (`SAP_CORE_BC_UI_JOU_PC`) to the business roles of Joule users, if your release lists it; without it the launchpad may not show Joule. (SAP documents this catalog for IBP, which shares the S/4HANA Cloud foundation.) VERIFY
9. Make sure test users have business roles whose apps are exposed; for Joule for Developers add catalog `SAP_A4C_BC_DEV_AIQ_PC` (allow up to 24 hours).
10. Sign out and in again (or use a private window), open the Fiori launchpad and check the Joule icon.

> **Source**
> SAP Help: Integration with SAP S/4HANA Cloud Public Edition; SAP Community: Joule in SAP S/4HANA Cloud Public Edition – Setup Guide.

8.2 SAP S/4HANA Cloud Private Edition (RISE)

Prerequisites

- RISE with SAP contract; S/4HANA release **2021 or later**, with **2023 FPS01** recommended.
- SAP Cloud Connector connected to the Joule subaccount; principal propagation configured.
- Fiori launchpad users authenticate through the shared IAS tenant.
- SAP Note `3639572` (Enabling Joule with SAP S/4HANA Cloud Private Edition) read and applied. VERIFY

Steps (SAP Help order)

1. Establish trust between the subaccount and IAS (7.4).

2. Configure trusted domains in BTP and IAS, including the S/4HANA launchpad host (7.5).
3. Configure user attributes so S/4HANA users carry the Global User ID (7.6).
4. Expose Fiori launchpad content: activate the content-exposure HTTP services in transaction SICF, either `/sap/bc/ui2/cdm3` (version 1) or `/sap/bc/http/ui2/flp_content_exposure/entities` (version 2, from 2023 FPS01). Then run `/UI2/CDM3_EXP_SCOPE` to select the roles to expose.
5. Check exposure logs in `/UI2/CDM3_EXP_SCOPE` (View Log) or transaction SLG1, object `/UI2/BE`, sub-object `/UI2/CDM3`.
6. In Cloud Connector, map a virtual host to the S/4HANA host (HTTPS) and expose the accessible resources with all sub-paths: the content-exposure path from the previous step, `/sap/opu/odata/sap/` (and `/sap/opu/odata4/` if your scenarios use OData V4), and `/sap/bc/ui2/`. VERIFY
7. Configure principal propagation: in Cloud Connector trust the subaccount and create the system certificate; in S/4HANA import the Cloud Connector CA certificate in transaction **STRUST** and create a rule-based certificate mapping in transaction **CERTRULE** from the certificate subject to the ABAP user.
8. Test that calls arrive as the named user, not a technical user (for example with the Cloud Connector connection check and **SU53** for an authorisation failure).
9. Create destinations: a design-time destination (content exposure, basic authentication) and a runtime destination (principal propagation) for Work Zone.
10. In Work Zone, open **Channel Manager → New** and add the S/4HANA system as a content provider; fetch content.
11. Run the booster (7.8) for the Private Edition scenario and check the destinations it creates.
12. Configure IPS so S/4HANA roles and users reach Work Zone (7.10).
13. Enable Joule in the S/4HANA Fiori launchpad as described in SAP Note 3639572. VERIFY
14. Schedule a regular content-update job so launchpad content stays in sync.
15. Test with real business roles (section 11).

> **Automation and multi-system**
> The **Cloud Integration Automation service** can drive much of this provisioning for Private Edition.
> SAP published multi-system Joule support for Private Edition in 2026, for landscapes with more than one S/4HANA system; Work Zone acts as the central launchpad that combines their content, each system through its own Cloud Connector virtual host. VERIFY
> Related notes: 3523238 (semantic objects and actions list), 3641945 (Joule icon displayed wrongly in the launchpad).

8.3 SAP S/4HANA on-premise and SAP ECC

Until 2026 SAP did not offer native Joule for on-premise S/4HANA or ECC; the documented route was a hybrid setup through BTP and Cloud Connector, mainly for proofs of concept. At SAP Sapphire 2026, SAP announced that **Joule Work** and a share of Joule assistants and agents would work in hybrid landscapes with on-premise S/4HANA and ECC, tied to a cloud commitment (sources mention RISE and the Max Success Plan). VERIFY

1. Ask SAP whether your contract qualifies and which release levels are supported.
2. If it does, follow the Private Edition pattern (8.2): IAS trust, Global User ID, content exposure, Cloud Connector with principal propagation, Work Zone content provider, booster.
3. Treat SAP Note 3632703 (cited for the earlier on-premise position) and SAP's current announcement materials as the authority. VERIFY
4. Expect less for ECC than for S/4HANA: ECC has no Fiori content exposure, so launchpad-style navigation and transactional scenarios depend entirely on what SAP's hybrid offering provides. Confirm the scenario list for ECC before committing to a scope.

8.4 SAP SuccessFactors

Prerequisites

- Valid SuccessFactors licence on a data centre that supports Joule.
- SuccessFactors authenticating through SAP Cloud Identity Services (usually delivered with the SuccessFactors tenant), with IPS in use.
- SAP Build Work Zone, standard edition / SAP Start available.
- SAP Discovery Center mission **Activate Joule for SAP SuccessFactors** for guided steps.

Steps

1. Confirm Cloud Identity Services is active and establish trust between it and the subaccount (7.4). SAP states this is required for the updated booster.
2. In IAS, add the SuccessFactors domain as a trusted domain.
3. Run the booster (7.8) and select the SuccessFactors system.
4. In Cloud Identity Services, add a source system, import the Joule JSON file for SuccessFactors, and rename the system to your SuccessFactors company ID. VERIFY
5. In SuccessFactors **Admin Center → Manage Permission Roles**, grant **General User Permission → Access to Joule** to the roles that should use Joule, and assign those roles to groups.
6. Provision the groups to Work Zone with IPS (7.10).
7. Sign in as a test user and check the Joule button. If it is missing for some users, see KBA 3685621.
8. Use the SuccessFactors company ID exactly as it is spelled (case included) wherever the setup asks for it. VERIFY
9. Joule runs embedded in SuccessFactors, so browser policies that block third-party cookies can hide it. Check your managed-browser policy, and ask SAP whether an IAS custom domain is recommended for your landscape.

> **Extension agents**
> Custom agents that read SuccessFactors data should propagate the user's identity so role-based permissions apply, rather than running on technical-user credentials.

8.5 SAP Concur

1. Configure trust to IAS and trusted domains in BTP and IAS (7.4, 7.5).
2. Configure user attributes from the Identity Directory (7.6).
3. Run the booster and select the Concur system. For an existing formation, use **Include Systems**.
4. Check the formation status is **Ready** and the Joule tenant appears in the systems landscape.
5. Add the Concur domain under the security settings as the post-integration step.
6. Assign users and test.

8.6 SAP Ariba

1. Make sure you have the Ariba Organization Admin role and know your realm's data centre.
2. Create the Ariba applications that generate the OAuth client ID and secret (in the SAP Ariba developer portal, with the procurement and sourcing APIs the scenarios need; the realm administrator approves them). VERIFY
3. Configure single sign-on through SAP Cloud Identity Services.
4. Run the booster and choose Ariba Buyer and Ariba Sourcing systems.
5. Assign users and test.

> **Source**
> SAP Community: Joule Integration with SAP Ariba – Setup Guide. Confirm the current steps on the SAP Help Ariba integration page. VERIFY

8.7 SAP Analytics Cloud

- SAP Build Work Zone is required (since Joule 2506): it provisions and syncs SAC users and roles; SAC roles and teams must sync to Work Zone as content.
- SAC authenticates with SAP Cloud Identity Services; keep at least **three free OAuth client slots**.
- Only one SAC tenant per Joule formation.
- Users need **Read** on the Joule privilege; enable **Just Ask** under the Conversational Analytics settings (check under System Administration → Default Appearance).
- For analytical insights, the user's SAC role needs read access to the content.
- Joule queries SAC as the user, so SAC data access control applies: test with a restricted user and expect answers limited to what that user can see.

8.8 SAP Datasphere

- Cloud Foundry subaccount; subscription-based, CPEA or SAP-managed Datasphere tenant with at least **one free OAuth client slot**.
- Users need role **DW Joule Consumer** (or a role with the Joule privilege); the tenant administrator activates Joule.
- The subaccount XSUAA subject name identifier must match the Datasphere identity provider setting.
- Add Datasphere trusted domains to the Joule IAS; synchronise users, roles and permissions to Work Zone with IPS.

8.9 SAP Integrated Business Planning

1. Register the IBP system with SAP BTP.
2. Run the booster with the IBP system and complete the IAS trust.
3. Add business catalog **User Interface – Joule Plugin** (`SAP_CORE_BC_UI_JOU_PC`) to the users' business roles.
4. Test with planners' roles.

8.10 Other SAP products

Product	Where to start
SAP Signavio solutions	SAP Help: Integration with SAP Signavio Solutions
SAP Fieldglass	SAP Help: Integrating with SAP Fieldglass
SAP Integrated Product Development	SAP Community: Joule Setup for SAP Integrated Product Development
SAP Sales Cloud / Service Cloud Version 2 and other CX	SAP Help integration page for the product VERIFY
SAP Integration Suite, SAP Build	SAP Help integration pages; Joule features are often enabled inside the product VERIFY
Unified setup across several products	SAP Community: Joule end-to-end setup guide for all lines of business (unified approach); SAP Learning: Setting Up a Unified Joule Instance

9. Extending Joule

9.1 Document grounding (answers from your own documents)

Grounding lets Joule answer from company documents. Documents are turned into vector embeddings by the Document Grounding service on SAP BTP and stored in SAP HANA Cloud Vector Engine.

1. In the Joule subaccount (or as SAP's guide directs), add the **Document Grounding** entitlement (plan `data-manager`), create the instance and a service key.
2. For Microsoft SharePoint: register an app in Microsoft Entra ID, grant it **Sites.Selected** (least privilege, preferred over Sites.Read.All), grant it access to the specific sites, and record the client ID and secret.
3. Create a pipeline for each source: SharePoint, SAP Document Management Service (via a BTP destination) or Google Drive.
4. Run the pipeline and check its status; then enable grounding for Joule as the guide describes.
5. Optionally add metadata ("contextualization") so answers can be filtered, and use the Cloud Integration Automation service to automate setup.
6. Test with questions whose answers exist only in the documents.

> **Document hygiene**
> Do not assume personal data is masked during indexing. Keep sensitive or personal data out of the folders you connect, or filter them first.
> Indexing runs as pipeline jobs, so new or changed documents appear after the next run, not instantly. VERIFY
> Check SAP Help for the supported file types and size limits before choosing sources. VERIFY

9.2 Microsoft 365 Copilot and Microsoft Teams

1. Federate SAP Cloud Identity Services with Microsoft Entra ID using **OpenID Connect** (SAML is only suitable when Entra is used purely for login).
2. Configure user mapping so Entra and IAS users match.
3. Register the SAP Joule application in Entra ID; grant admin consent for the SAP Copilot API access app, then under the federation app's **API permissions** add `svc-ai-joule-copilot-api-access` with delegated `access_as_user` and grant consent. VERIFY
4. In BTP System Landscape, add a system of type **Microsoft 365 Copilot for SAP Joule** and include it in the Joule formation.
5. Enable the Joule agent in the Microsoft 365 admin centre or Teams admin centre.
6. Use a separate Entra ID tenant per Joule environment (a single Entra tenant cannot serve several Joule environments), and in any case a separate app registration for non-production and production.
7. Test both directions: Copilot/Teams passing requests to Joule, and Joule passing requests to Copilot.

9.3 Joule Studio (custom skills and agents)

1. Entitlements: SAP Build Process Automation with plan **`build_default`** (tenants on plan `standard` are not compatible), Joule, and an SAP Cloud Identity Services tenant.
2. Configure OIDC trust to IAS and disable the default identity provider for login.
3. Run the Joule booster from the global account if Joule is not yet set up.
4. Create role collections: a Joule admin collection with the Joule roles, and `ProcessAutomationAdmin` (plus developer roles) for builders; assign to your IAS users or groups.

5. Configure destinations to back-end systems the skills will call, with the additional property `sap.processautomation.enabled = true` so SAP Build can use them; prefer principal propagation over technical users.
6. Build a skill or agent in SAP Build (Joule Studio), test it, and deploy it to the Joule tenant; use the SAP Build lifecycle (with SAP Cloud Transport Management where used) to promote from test to production.
7. Follow the SAP Discovery Center mission **Setup Joule Studio for a BTP Enterprise Account**, or SAP-managed provisioning for Joule Studio where available.

9.4 Joule Agents

- SAP-delivered Joule Agents are premium capabilities that consume AI Units.
- Administrators grant users permission to specific agents through the product's roles (for example SuccessFactors agents through role-based permissions).
- Plan consumption in advance and monitor it in SAP for Me and the Joule Analytics Center (12.2).

10. Joule for Consultants and Joule for Developers

10.1 SAP Joule for Consultants

- Admins needed: BTP global account administrator, SAP Cloud Identity Services administrator, and Cloud Administrator or Super Administrator in SAP for Me.
- Use a **dedicated subaccount**; integrating with another Joule instance in the same subaccount does not work.
- Supported regions cited: EU10, EU11, EU20, US10, AP10, JP10. VERIFY

1. Assign entitlements: Joule (`foundation`) and **SAP Consulting Capability** (`standard ga`).
2. Activate the service from the subscription; check **Instances and Subscriptions**.
3. Create a role collection with `end_user` from `das-application`, for example `JOULE_FOR_CONSULTANTS_END_USER-<subaccount>`.
4. Assign users through an IAS group mapped to the role collection, or directly.
5. Open the console from **Instances and Subscriptions**. Optionally add custom knowledge grounding.

10.2 SAP Joule for Developers (ABAP AI capabilities)

System	Setup
S/4HANA Cloud Public Edition / BTP ABAP environment	Assign users a business role containing catalog `SAP_A4C_BC_DEV_AIQ_PC`; allow up to 24 hours.
S/4HANA Cloud Private Edition 2025 (newer posts: 2021 and higher)	Subscribe to the service in BTP; generate a client certificate; configure certificate mapping in S/4HANA (Maintain Certificate Mapping app, IAS group required); assign authorisation object `S_AIQADTLO`; allowlist the regional proxy URL. SAP Note `3558794` covers known issues. VERIFY

11. Testing and acceptance

Test in the non-production tier with real business roles, then repeat the core tests in production before go-live.

ID	Test	How	Expected result
T01	Single sign-on	Open the product as a test user in a private window and launch Joule	Joule opens without a second login
T02	Navigation	"Open Manage Purchase Orders" (or a common app in the product)	Joule offers the app the user is allowed to open
T03	Informational	"What is a credit memo?" / "How do I request leave?"	Grounded answer from SAP help content
T04	Transactional	"Show my open purchase orders" / "Show my team's time off"	Live data from the product, limited to the user's permissions
T05	Authorisation (negative)	Repeat T04 with a user lacking the business role	Joule declines or shows no data
T06	Cross-product	Ask something that needs a second integrated product	Joule routes to the right product
T07	Document grounding	Ask about content only in your grounded documents	Answer cites the document
T08	Microsoft 365	Ask Joule from Teams / Copilot, and Copilot from Joule	Requests pass both ways
T09	Custom skill or agent	Trigger the Joule Studio skill	Skill runs with the user's identity
T10	Usage analytics	Open the Joule Analytics Center after tests	Test sessions appear by product and scenario
T11	Mobile / other clients	Use Joule from the supported mobile or desktop client if in scope	Same behaviour as browser

11.1 Diagnostic tips while testing

- **T01 login problems:** record the sign-in in the browser developer tools (Network tab). A healthy flow redirects once between Joule and IAS and back; repeated redirects point to trust or subject-name settings (7.4, 7.6).
- **Check the token:** decode the user's token with a local tool and confirm `user_uuid` matches the user's Global User ID in IAS and that the expected groups are present. Do not paste production tokens into public websites.
- **T04 and T05:** if a user without the business role still gets data, stop and review the product-side authorisation setup before go-live.

> **Exit criteria**
> All tests pass for every product in scope, no admin roles are assigned to test end users, analytics show the sessions, and open issues are logged with SAP (component `CA-JOULE`) where needed.

12. Operating Joule

12.1 Administration

- **Joule Admin Center:** `<joule-tenant-url>/admin`, role `tenant_admin`.
- **Role management:** one shared role collection for all Joule roles, or separate ones for finer control; review quarterly.
- **User provisioning:** keep IPS jobs scheduled; leavers must be removed in IAS so Joule access ends.
- **Release cadence:** Joule updates frequently; validate new capabilities on the Joule Preview (non-production) instance before they reach production.

12.2 Usage and cost monitoring

- **Joule Analytics Center:** `<joule-tenant-url>/analytics`, role `analytics_admin`. Shows product usage, scenario usage, interaction-type usage and client-type usage. SAP describes it as the dedicated source of Joule usage; usage is not shown in SAP for Me.
- **AI Units:** track balance and consumption in SAP for Me; agree alert thresholds with Finance.

12.3 Support

- Open SAP incidents under component `CA-JOULE` (product-specific components may also apply).
- Keep a runbook of your formation name, subaccounts, IAS tenants, destinations and role collections for each tier.

12.4 Certificate lifecycle

Private Edition and on-premise integrations depend on certificates: the Cloud Connector system and CA certificates, the S/4HANA trust entries (STRUST), destination certificates and identity-provider signing certificates. When one expires, Joule stops working for everyone at once.

- List every certificate with its expiry date in your service-management or ALM tool.
- Set reminders well before expiry (for example 60, 30 and 7 days).
- Renew and import the new certificate in every place that trusts it before the old one expires, then test.

13. Troubleshooting

Symptom	Likely cause	Fix
Booster stops at the Work Zone check	SAP Build Work Zone not subscribed or not registered in the subaccount	Subscribe to Work Zone, standard edition in the Joule subaccount; see KBA `3721340`
Booster reports no Identity Authentication tenant	The global account has no IAS tenant linked	Link or create the Cloud Identity Services tenant; see KBA `3675105`
Joule icon missing for everyone	Role collection not assigned; Content Security Policy or trusted domain missing; Joule not enabled in the launchpad; browser cache	Check 7.5 and 7.9; for S/4HANA Public re-check "Update Content Security Policy Settings"; try a private window and clear cache
Joule works for admins, not for users	Users lack the `end_user` role collection or the IAS group mapping	Map the role collection to the users' IAS group (7.9)
Joule button missing for some SuccessFactors users	Missing General User Permission → Access to Joule; roles not synced; browser or network policy blocks URLs	Follow KBA `3685621`: permissions, sync, third-party cookies, allowlist SuccessFactors, IAS and Joule URLs
Login loop or session breaks after Joule opens	Different IAS tenants, SAML trust, or subject name identifier not the Global User ID in Joule or Work Zone	Use one IAS tenant, OIDC trust, Global User ID everywhere (7.4, 7.6)
"App not found" or wrong app on navigation	Work Zone content not synced, roles not provisioned, or (Private Edition) content not exposed	Refresh the content provider, run IPS, check `/UI2/CDM3_EXP_SCOPE` and SLG1
Private Edition content provider fails	SICF service inactive, Cloud Connector path missing, wrong destination	Activate the content-exposure service, expose the path with sub-paths, check the design-time destination (8.2)
Joule icon displayed wrongly in the Private Edition launchpad	Known issue	SAP Note `3641945`
SAC "Just Ask" not offered in Joule	Privilege or feature not enabled, no free OAuth clients, or second SAC tenant in the formation	Follow 8.7
Grounding returns nothing	Pipeline failed or not run; SharePoint consent missing	Check pipeline status; grant Sites.Selected access to the site (9.1)
Teams / Copilot cannot reach Joule	Entra–IAS trust on SAML, consent missing, Entra tenant reused across environments	Use OIDC, grant `svc-ai-joule-copilot-api-access`, one Entra tenant per environment (9.2)
Joule not supported in ADT (Private Edition)	Known issue for ABAP AI capabilities	SAP Note `3558794`
Unexpected AI Unit consumption	Agents or premium features used more than planned	Review the Joule Analytics Center and SAP for Me; restrict agent permissions to the users who need them
Users have no role collections although mappings exist	The IAS application does not send groups in the token	Send the `groups` attribute (7.6); sign out and in again

Symptom	Likely cause	Fix
HTTP 403 inside Joule right after login	User signed in through SAP ID Service instead of IAS	Clear Available for User Logon on the default identity provider (7.4)
System missing from the booster	System is only discovered, not registered	Register it in System Landscape and complete the product-side step (7.7)
"System can be part of only one formation"	System already belongs to an earlier or half-built formation	Reuse or clean up that formation before running the booster again (7.8)
Booster cannot be started	Your user lacks global-account (or System Landscape) administration rights	Get the roles in 5.1
Work Zone content provider "sync failed"	Design-time destination has wrong credentials or path	Check the design-time destination and Cloud Connector path (8.2)
Duplicate users in IAS or Work Zone	Provisioning keyed on e-mail instead of the Global User ID	Re-key IPS on the immutable ID and clean up duplicates (7.10)
Joule suddenly fails for everyone (Private Edition)	Expired Cloud Connector or trust certificate	Renew and import the certificate; see 12.4
Slow responses	Joule subaccount far from the back-end region	Prefer a co-located region for the Joule subaccount (4.2)
Transactional answers show data a user should not see	Product-side authorisation or communication setup too broad	Stop rollout; review roles and communication arrangements with the product team
Joule missing in S/4HANA Cloud for some roles	Joule plugin catalog not in the business role	Add `SAP_CORE_BC_UI_JOU_PC` where your release lists it (8.1)
Teams / Copilot "Access denied"	Admin consent missing for the delegated permission	Grant admin consent in Entra ID (9.2)

> **Escalation**
> If the fixes above do not help, open an incident under `CA-JOULE` with the formation name, subaccount ID, IAS tenant, product system, user, time of the attempt and a HAR file from the browser.

14. Security and data protection

- Joule acts with the signed-in user's identity; product authorisations decide what data it can show or change. Keep role design and segregation of duties current.
- SAP states it does not share customer data with third-party model providers to train their models, and that prompts sent to third-party hosted models are processed, not retained. Model routing is SAP-managed through the generative AI hub in SAP AI Core. VERIFY
- Confirm data residency and retention for your regions in the Joule security guide and SAP's data-protection documentation before production. VERIFY
- Use least-privilege access for grounding sources (for example SharePoint Sites.Selected) and review which documents are indexed.
- For custom agents, propagate user identity rather than using technical users, so permissions apply.
- Include Joule in your AI governance: acceptable use, human review of agent actions, audit of admin role assignments, and incident handling.

14.1 Security checklist

☐	Item	Owner	Ref
☐	Data-protection impact assessment completed	DPO	14
☐	Residency and retention confirmed for chosen regions	Security	14
☐	Admin role collections limited to named administrators	BTP admin	7.9
☐	Grounding sources reviewed; least-privilege access	Content owner	9.1
☐	Custom skills use principal propagation / user identity	Developers	9.3
☐	User communication and acceptable-use guidance published	Change lead	15

15. Master go-live checklist

Phase 1: Prepare

☐	Item	Owner	Ref
☐	Licences, Joule Base and AI Units confirmed	Project lead	3
☐	Provisioning model and landscape (tiers, regions, IAS tenants) approved	Architect	2, 4
☐	People and access in place	Project lead	5
☐	Global User ID present for all users	Identity admin	7.6

Phase 2: Build (non-production)

☐	Item	Owner	Ref
☐	SAP-managed provisioning complete, or customer-managed foundation complete	BTP admin	6 / 7
☐	Each product integrated and product-side settings done	Product admins	8
☐	Extensions configured (grounding, Microsoft 365, Joule Studio) if in scope	BTP admin	9
☐	Tests T01–T11 passed	Key users	11

Phase 3: Go live (production)

☐	Item	Owner	Ref
☐	Production tier built with the same steps	BTP admin	6 / 7 / 8
☐	Core tests repeated in production	Key users	11
☐	Role collections mapped to production IAS groups	Identity admin	7.9
☐	Users informed; quick-start guide and support route published	Change lead	12.3
☐	Analytics and AI Unit monitoring switched on	Joule admin	12.2

Phase 4: Operate

☐	Item	Owner	Ref
☐	Monthly review of usage, AI Units and incidents	Joule admin	12
☐	New Joule releases validated on Joule Preview first	Joule admin	12.1
☐	Quarterly role and access review	Security	14

Appendix A. Glossary

Term	Meaning
AI Units	SAP's consumption currency for premium AI capabilities such as Joule Agents.
Booster	Guided automation in the BTP cockpit; "Setting Up Joule" configures Joule in a subaccount.
CIAS	Cloud Integration Automation service: automates multi-step provisioning (Private Edition, grounding).
Content exposure (CDM3)	How S/4HANA Private Edition / on-premise publishes launchpad content to Work Zone.
Formation	A BTP System Landscape grouping of Joule and the systems it integrates.
Global User ID	The person-level identifier (`user_uuid`) that must match across IAS and all products.
IAS / IPS	Identity Authentication and Identity Provisioning, the two parts of SAP Cloud Identity Services.
Joule Base / Premium	Included Joule capabilities vs AI-Unit-based premium capabilities.
Joule Preview	SAP's preview landscape for non-production Joule tenants.
Joule Studio	The SAP Build capability for custom Joule skills and agents.
Principal propagation	Passing the signed-in user's identity from BTP to a private back end through Cloud Connector.
Work Zone, standard edition	SAP Build Work Zone; Joule uses its navigation service to find apps.

Appendix B. SAP Notes and Knowledge Base Articles

Open these in SAP for Me (support.sap.com) before setup. Titles are as cited in research; confirm the current version.

Number	Topic
2791410	SAP Cloud Identity Services integration: required configuration and considerations (cited for SuccessFactors)
3523238	Semantic objects and actions supported by Joule (spreadsheet)
3558794	AI functionality in S/4HANA Cloud Private Edition (Joule for Developers / ADT)
3632703	Joule and on-premise S/4HANA (earlier position) VERIFY
3639572	Enabling Joule with SAP S/4HANA Cloud Private Edition
3641945	Joule icon displayed wrongly in the Fiori launchpad of S/4HANA Cloud Private Edition
3675105	Joule booster: global account without an Identity Authentication tenant
3685621	Joule button not appearing on SuccessFactors
3721340	Joule booster: SAP Build Work Zone registration missing
3721491	Joule availability question (on-premise) VERIFY
3729978	Joule setup (SuccessFactors context) VERIFY

Appendix C. Key sources

The full list with confidence ratings is in `research-notes.md`. Start with these SAP pages:

- Joule Integration Guide (SAP Help, PDF): https://help.sap.com/doc/de3af3c0f81642dbaa4d36172ed57a72/CLOUD/en-US/79bfc83ab386450c8cd9c7937ce26a3a.pdf
- What is Joule (SAP Help, PDF, 2026-10-20): https://help.sap.com/doc/75f5965fbb514fbcb89f1e47c094ea3f/CLOUD/en-US/9d8c0ef2443b49f993a27de8f38f83af.pdf
- Joule Onboarding Guide: https://help.sap.com/docs/joule/serviceguide/onboarding-joule
- Customer-Managed Provisioning: https://help.sap.com/docs/joule/integrating-joule-with-sap/initial-setup
- Prerequisites for Customer-Managed Provisioning: https://help.sap.com/docs/joule/integrating-joule-with-sap/prerequisites
- Integration with S/4HANA Cloud Public Edition: https://help.sap.com/docs/joule/integrating-joule-with-sap/integration-with-sap-s-4hana-cloud-public-edition
- Integration with S/4HANA Cloud Private Edition: https://help.sap.com/docs/joule/integrating-joule-with-sap/integration-with-sap-s-4hana-cloud-private-edition
- Identity and Access Management for Joule: https://help.sap.com/docs/cloud-identity-services/cloud-identity-services/identity-and-access-management-for-joule
- Joule Landscape Recommendation (Architecture Center): https://architecture.learning.sap.com/docs/ref-arch/4c80fa
- Setting up Joule across your system landscapes (SAP Learning): https://learning.sap.com/courses/setting-up-joule-across-your-organizational-system-landscapes
- Joule end-to-end setup guide for all lines of business (SAP Community): https://community.sap.com/t5/technology-blog-posts-by-sap/joule-end-to-end-setup-guide-for-all-line-of-business-unified-approach/ba-p/13964602
- Managing Joule on SAP BTP (SAP Community): https://community.sap.com/t5/technology-blog-posts-by-sap/managing-joule-on-sap-btp-roles-analytics-administration-and/ba-p/14325933
- SAP Joule and Microsoft Copilot integration (Microsoft Learn): https://learn.microsoft.com/en-us/azure/sap/microsoft-ai/joule/joule-copilot-overview

Appendix D. Open items to verify

These details came from a single or non-SAP source, or sources disagreed. Close each one against SAP Help, SAP for Me or your SAP account team.

#	Item	Where used
1	Exact plan names for Joule (`foundation` vs `standard` / DAS Standard / Premium), Work Zone and Cloud Identity Services	7.2
2	SKU numbers and current AI Unit consumption rates	3
3	Supported regions for each Joule product and for Joule Preview	4.2, 10.1
4	Role names inside `das-application` for your data centre	7.9
5	S/4HANA Cloud Public Edition system registration steps	7.7, 8.1
6	Private Edition: Joule launchpad enablement per Note 3639572; multi-system support	8.2
7	On-premise / ECC eligibility, scope and dates after Sapphire 2026	8.3
8	SuccessFactors JSON source-system import in Cloud Identity Services	8.4
9	Ariba, CX, Integration Suite, Fieldglass, Signavio detailed steps	8.6, 8.10
10	Entra ID permission names for the Microsoft 365 integration	9.2
11	Joule for Developers on Private Edition: releases and authorisations	10.2
12	Data residency and retention statements	14
13	Joule plugin catalog `SAP_CORE_BC_UI_JOU_PC` for S/4HANA Cloud Public Edition	8.1
14	Discovered versus registered systems; one formation per system; booster resume behaviour	7.7, 7.8
15	System Landscape Administrator requirement	5.1
16	Cloud Connector resource paths for Joule scenarios in Private Edition	8.2
17	Ariba developer-portal application and scopes	8.6
18	Document grounding file types, limits and indexing schedule	9.1

Appendix E. Field notes to confirm

These specific values were suggested in the peer review without a cited source, and the original research did not find them. They may be right, but do not configure, budget or promise anything based on them until SAP documentation or your SAP contact confirms them.

Area	Suggested value	Confirm with
Commercial	`8019544` as the zero-cost Joule Base line item (one partner source also cites it)	Order form / SAP account team
Commercial	Joule for Consultants at 35 AI Units per user per month with a fair-use request cap	SAP account team
Commercial	Document grounding at 0.005 AI Units per indexed record, records sized by file size	SAP price list
Entitlements	Joule plan `preview` on the Preview landscape; Cloud Identity Services additional-tenant SKU	BTP cockpit entitlements
Regions	Productive Joule regions EU10, EU20, US10, US21, AP10, JP10; region-dependent image and slide processing for grounding	SAP Discovery Center
Booster	Seven booster screens, a 15–25 minute run, named error codes and destination names (for example `S4P_JOULE_DEST`)	The booster itself in your tenant
S/4HANA Cloud	Content Security Policy entries for specific Joule domains, including WebSocket	SAP Help: S/4HANA Cloud integration
Private Edition	Launchpad plugin component name and a launchpad configuration parameter to switch Joule on	SAP Note 3639572
Mobile	A dedicated mobile redirect URI in IAS	SAP Help: SAP Mobile Start / Joule
Joule Studio	A command-line tool for initialising, testing and deploying agents	SAP Help: Joule Studio
Joule for Developers	Minimum ADT version, field values of `S_AIQADTLO`, a model proxy host to allowlist, Private Edition release floor 2023 FPS01 (a newer SAP post says 2021 and higher)	SAP Note 3558794
SuccessFactors	Booster bindings that depend on the exact system name	SAP Help: SuccessFactors integration

Appendix F. Review suggestions not adopted

These peer-review suggestions were not adopted because they conflict with the SAP sources found, or because following them unverified could cost money, break identities or overstate compliance. They are listed so nobody acts on them by mistake.

Suggestion	Why not adopted
Reserve at least 18 GB of Cloud Foundry memory before running the booster	Joule is consumed as an SAP-run subscription; no SAP source found asks for your own runtime quota. Buying quota on this basis could waste money.
Switch communication arrangement `SAP_COM_0882` from an "EMPTY" to a "ROLE" mode	No such setting appears in the sources. Review authorisations with the product team instead (13).
Map SuccessFactors `personIdExternal` into the IAS `userUuid`	The Global User ID is managed by IAS; overwriting it can break sign-in across every connected product. SAP's guidance is to import the Joule source system as documented (8.4).
Treat on-premise S/4HANA 2021+ as fully eligible with CPEA and Joule Base	2026 reporting ties on-premise access to a cloud commitment; confirm eligibility with SAP (8.3).
State that EU data residency is guaranteed and prompts are never logged	These are legal statements; SAP sources say to confirm them in the Joule security and data-protection documentation (14).
Always choose customer-managed from day 1 if you want Joule Studio or grounding	SAP offers SAP-managed provisioning for Joule Studio and positions SAP-managed as the default; decide per extension (2.1).
Verify e-mail domains with DNS records or IPS quarantines users	Not found in the sources; IAS custom domains and e-mail verification are separate topics.
Paste tokens into a public JWT website to inspect them	Exposes live credentials; decode locally instead (11.1).